Inner product spaces 3

Find an orthonormal basis with the Gram-Schmidt process.

Existence of an orthonormal basis

Every nonzero finite-dimensional inner product space has an orthonormal basis.

Consequence: Every vector subspace (with the restriction of the inner product) has an orthonormal basis.

To construct an orthogonal or an orthonormal basis we apply the **Gram-Schmidt process** to a given basis.

Remark: The order of the given basis matters. Different orderings usually lead to different orthogonal basis (for example we always keep the first vector).

Gram-Schmidt process

Start with a basis $u_1, \ldots, u_r$. We construct an orthogonal basis $v_1, \ldots, v_r$ (which we can easily turn into an orthonormal basis by normalizing the vectors).

- $v_1 = u_1$
- $v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{||v_1||^2} v_1$
- $v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{||v_1||^2} v_1 - \frac{\langle u_3, v_2 \rangle}{||v_2||^2} v_2$
- $v_4 = u_4 - \frac{\langle u_4, v_1 \rangle}{||v_1||^2} v_1 - \frac{\langle u_4, v_2 \rangle}{||v_2||^2} v_2 - \frac{\langle u_4, v_3 \rangle}{||v_3||^2} v_3$
- $\ldots$
- $v_r = u_r - \sum_{i=1}^{r-1} \frac{\langle u_r, v_i \rangle}{||v_i||^2} v_i$

$$v_k = u_k - \sum_{i=1}^{k-1} \frac{\langle u_k, v_i \rangle}{||v_i||^2} v_i \quad \text{for all } k = 1, \ldots, r$$

Example

In $\mathbb{R}^3$ with the Euclidean inner product apply Gram-Schmidt to

$$\mathbf{u}_1 = (1,1,1) \qquad \mathbf{u}_2 = (0,1,1) \qquad \mathbf{u}_3 = (0,0,1)$$

We get an an orthogonal basis $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, which we then normalize to $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$.

$$\begin{aligned}
&\textit{Step 1.}\ \mathbf{v}_1 = \mathbf{u}_1 = (1,1,1)\\
&\textit{Step 2.}\ \mathbf{v}_2 = \mathbf{u}_2 - \operatorname{proj}_{W_1}\mathbf{u}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1\rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1\\
&\qquad = (0,1,1) - \frac{2}{3}(1,1,1) = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)\\
&\textit{Step 3.}\ \mathbf{v}_3 = \mathbf{u}_3 - \operatorname{proj}_{W_2}\mathbf{u}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1\rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2\rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2\\
&\qquad = (0,0,1) - \frac{1}{3}(1,1,1) - \frac{1/3}{2/3}\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)\\
&\qquad = \left(0, -\frac{1}{2}, \frac{1}{2}\right)
\end{aligned}$$

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad \mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right),$$

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \blacktriangleleft$$

Remarks on the example

In the previous example, had we chosen the ordering $\mathbf{u}_3, \mathbf{u}_2, \mathbf{u}_1$ we would have obtained the standard basis.

If the question is finding an orthonormal basis for $\mathbb{R}^3$ for the Euclidean standard product, we could have immediately written the standard basis. Sometimes it is possible to guess some nice vectors and extend them to an orthonormal basis (see later on).
However in general we need Gram-Schmidt if we handle a subspace of $\mathbb{R}^3$ or if we use a different inner product of $\mathbb{R}^3$.

Given an ordered basis, the result of Gram-Schmidt is determined. However we can simplify the calculations and rescale the vectors in intermediate steps (to get rid of denominators or huge common factors):

$$(\frac{1}{10}, \frac{2}{10}, \frac{3}{10}) \quad (10, 20, 30) \quad \rightsquigarrow (1, 2, 3)$$

Proof

We prove by induction on n $(1 \leqslant n \leqslant r)$ that the sets $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are linearly independent, generate the same subspace, and $v_1, \ldots, v_n$ is an orthogonal set.

- The set $u_1, \ldots, u_n$ is a subset of $u_1, \ldots, u_r$ hence it is also linearly independent; the set $v_1, \ldots, v_n$ will be linearly independent as a consequence of the fact that it has the same cardinality of a linearly independent set which generates the same subspace.
- By induction: $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ generate the same subspace. For $n = 1$ this is clear because $u_1 = v_1$. For $n \rightsquigarrow n + 1$

$$v_{n+1} = u_{n+1} - \sum_{i=1}^{n} \frac{\langle u_n, v_i \rangle}{||v_i||^2} v_i$$

so $v_{n+1} - u_{n+1}$ is in $Span(v_1, \ldots, v_n) = Span(u_1, \ldots, u_n)$. Thus adding v_{n+1} gives us the same vectors as adding u_{n+1}.

Proof

- By induction: the set $v_1, \ldots, v_n$ is orthogonal. For $n = 1$ it is trivial. For $n \rightsquigarrow n+1$, let $1 \leqslant k \leqslant n$ and consider

$$v_{n+1} = u_{n+1} - \sum_{i=1}^{n} \frac{\langle u_{n+1}, v_i \rangle}{||v_i||^2} v_i$$

We get:

$$\langle v_k, v_{n+1} \rangle = \langle v_k, u_{n+1} \rangle + \langle v_k, -\sum_{i=1}^{n} \frac{\langle u_{n+1}, v_i \rangle}{||v_i||^2} v_i \rangle$$

$$= \langle v_k, u_{n+1} \rangle + \langle v_k, -\frac{\langle u_{n+1}, v_k \rangle}{||v_k||^2} v_k \rangle$$

$$= \langle v_k, u_{n+1} \rangle - \frac{\langle u_{n+1}, v_k \rangle}{||v_k||^2} \langle v_k, v_k \rangle = 0$$

Extending Orthogonal Sets to Orthogonal Bases

Let W be a finite-dimensional inner product space.

- Every orthogonal (orthonormal) set of nonzero vectors in W can be enlarged to an orthogonal (orthonormal) basis for W .

Proof. The orthogonal set of nonzero vectors $v_1, \ldots, v_s$ is a linearly independent set, so that it can be enlarged to a basis. Apply the Gram–Schmidt process to this basis. For every $1 \leqslant k \leqslant s$ by induction

$$v_{k,new} = v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, v_i \rangle}{||v_i||^2} v_i = v_k$$

The Gram–Schmidt process gives an orthogonal basis where the first vectors are $v_1, \ldots, v_s$. If we normalize the vectors we get an orthonormal basis (and the first vectors are $v_1, \ldots, v_s$, if these were an orthonormal set).

Decompose a matrix with independent column vectors.

QR-Decomposition

If A is an $m \times n$ matrix with linearly independent column vectors, then A can be factored as

$$A = QR$$

where

- Q is an $m \times n$ matrix with orthonormal column vectors
- R is an $n \times n$ invertible upper triangular matrix.

Remark: The result applies in particular to an invertible square matrix.

Proof of the QR-Decomposition

$$A = \begin{pmatrix} u_1 | & u_2 | & \dots | & u_n \end{pmatrix} \qquad Q = \begin{pmatrix} q_1 | & q_2 | & \dots | & q_n \end{pmatrix}$$

The column vectors of Q are obtained from those of A by applying Gram-Schmidt (in particular, A and Q have the same column space).

For every $1 \leqslant k \leqslant n$

$$u_k = \langle u_k, q_1 \rangle q_1 + \cdots + \langle u_k, q_n \rangle q_n$$

and hence $A = QR$ for some R because

$$A = \begin{pmatrix} q_1 | & q_2 | & \dots | & q_n \end{pmatrix} \begin{pmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \cdots & \langle u_n, q_1 \rangle \\ \langle u_1, q_2 \rangle & \langle u_2, q_2 \rangle & \cdots & \langle u_n, q_2 \rangle \\ \vdots & \vdots & & \vdots \\ \langle u_1, q_n \rangle & \langle u_2, q_n \rangle & \cdots & \langle u_n, q_n \rangle \end{pmatrix}$$

Proof of the QR-Decomposition

The $n \times n$ matrix R is upper triangular if

$$\langle u_k, q_j \rangle = 0 \qquad \forall k < j$$

This is true because by Gram-Schmidt

$$q_k = u_k - \sum_{i=1}^{k-1} \frac{\langle u_k, q_i \rangle}{||q_i||^2} q_i$$

$$u_k = q_k + \sum_{i=1}^{k-1} \frac{\langle u_k, q_i \rangle}{||q_i||^2} q_i$$

thus

$$u_k \in Span(q_1, \ldots, q_k)$$

We conclude because q_j is orthogonal to $q_1, \ldots, q_k$ and hence also to all their linear combinations.

Proof of the QR-Decomposition

The matrix R is then invertible if the entries on the diagonal are non-zero:

$$\langle u_j, q_j \rangle \neq 0 \qquad \forall j = 1, \ldots, n$$

This is true because

$$u_j = q_j - \sum_{i=1}^{j-1} \frac{\langle u_j, q_i \rangle}{||q_i||^2} q_i$$

$$\langle u_j, q_j \rangle = \langle q_j, q_j \rangle - \sum_{i=1}^{j-1} \frac{\langle u_j, q_i \rangle}{||q_i||^2} \langle q_j, q_i \rangle = \langle q_j, q_j \rangle = ||q_j||^2 \neq 0$$

Example

Find the QR decomposition of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

Calling $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ the column vectors, to find Q we apply Gram-Schmidt, while

$$R = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \langle \mathbf{u}_3, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \langle \mathbf{u}_3, \mathbf{q}_2 \rangle \\ 0 & 0 & \langle \mathbf{u}_3, \mathbf{q}_3 \rangle \end{bmatrix} = \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

from which it follows that a QR-decomposition of A is

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}}_{R}$$

Inventing solutions for inconsistent systems.

Preliminary: Best Approximation Theorem

Let W be a finite-dimensional subspace of an inner product space V. Let b be a vector in V.
The best approximation to b from W is $proj_W b$. This means

$$||b - w|| \geq ||b - proj_W b|| \qquad \forall\, w \in W$$

Proof.

$$b - w = (b - proj_W b) + (proj_W b - w)$$

Since $b - proj_W b$ is orthogonal to W the two terms on the RHS are orthogonal and by Pythagora's Theorem

$$\underbrace{||b - w||^2}_{\text{to be minimized}} = \underbrace{||b - proj_W b||^2}_{\text{fixed}} + \underbrace{||proj_W b - w||^2}_{\text{minimal when zero}}$$

Best approximation

Suppose that

$$Ax = b$$

is an inconsistent linear system of m equations in n unknowns.
Maybe the inconsistency is caused by errors in the entries of A or b (the correct entries should be minor variations of the given ones).
Since no exact solution is possible, we look for a vector x that is "as close as possible" to being a solution.
Being a solution would mean $b - Ax = 0$ or, equivalently,

$$||b - Ax|| = 0$$

so we now require that x minimizes $||b - Ax||$.
The norm comes from the Euclidean inner product on $\mathbb{R}^m$.

Least Squares Problem

Given a linear system $Ax = b$ of m equations in n unknowns, find a vector $x \in \mathbb{R}^n$ that minimizes $||b - Ax||$ with respect to the Euclidean inner product on $\mathbb{R}^m$.

x is a **least squares solution**
$b - Ax$ is the **least squares error vector**
$||b - Ax||$ is the **least squares error**.

Since $\{Ax : x \in \mathbb{R}^n\} = col(A)$ we are looking for the vector in $col(A)$ which is closest to b. This vector is $proj_{col(A)}b$.
The least squares error vector (and hence also the least square error) is unique, being $b - proj_{col(A)}b$. The least square solutions are in general not unique (there could be infinitely many x such that $Ax = proj_{col(A)}b$).

Remark: If $Ax = b$ would be consistent, the least square error is zero and the least square solutions are precisely the solutions.

Remark: We speak of least squares because

$$||b - Ax||^2 = (b - Ax)_1^2 + \cdots + (b - Ax)_m^2$$

Best Approximation Theorem

Recall that the least square error vector, which is unique, is

$$b - proj_{col(A)} b$$

To compute it, we can simply compute the projection (compute an orthonormal basis of $col(A)$ first, and then apply the projection formula).

To compute the least square error, simply compute the norm of the least square error vector.

To compute the least square solutions one simply has to solve the linear system

$$Ax = proj_{col(A)} b$$

Key point: this linear system is solvable, as $proj_{col(A)} b \in col(A)$. There can be infinitely many least square solutions (this depends on A).

Finding least square solutions (with the normal system)

The normal system associated to $Ax = b$ is

$$A^T Ax = A^T b$$

Its solutions x are the least squares solutions of $Ax = b$ (in particular, it is consistent).

Proof: Rewrite the system as $A^T(b - Ax) = 0$.
Recall that $col(A)^\perp = null(A^T)$.

$Ax = proj_{col(A)} b$ implies

$$A^T(b - Ax) = A^T(b - proj_{col(A)} b) = 0$$

because $b - proj_{col(A)} b \in col(A)^\perp = null(A^T)$.

Conversely if $A^T(b - Ax) = 0$ we deduce that
$b - Ax \in null(A^T) = col(A)^\perp$ hence by the Projection Theorem
$Ax = proj_{col(A)} b$.

Example

Solution It will be convenient to express the system in the matrix form $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} \tag{7}$$

It follows that

$$A^T A = \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 14 & -3 \\ -3 & 21 \end{bmatrix} \tag{8}$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

so the normal system $A^T A\mathbf{x} = A^T \mathbf{b}$ is

$$\begin{bmatrix} 14 & -3 \\ -3 & 21 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

Solving this system yields a unique least squares solution, namely,

$$x_1 = \tfrac{17}{95}, \quad x_2 = \tfrac{143}{285}$$

The least squares error vector is

$$\mathbf{b} - A\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} \frac{17}{95} \\ \frac{143}{285} \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} -\frac{92}{285} \\ \frac{439}{285} \\ \frac{95}{57} \end{bmatrix} = \begin{bmatrix} \frac{1232}{285} \\ -\frac{154}{285} \\ \frac{4}{3} \end{bmatrix}$$

and the least squares error is

$$\|\mathbf{b} - A\mathbf{x}\| \approx 4.556$$

Example

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & -4 & 3 \\ 1 & 10 & -7 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

It follows that

$$A^T A = \begin{bmatrix} 11 & 12 & -7 \\ 12 & 120 & -84 \\ -7 & -84 & 59 \end{bmatrix} \quad \text{and} \quad A^T \mathbf{b} = \begin{bmatrix} 5 \\ 22 \\ -15 \end{bmatrix}$$

so the augmented matrix for the normal system $A^T A\mathbf{x} = A^T \mathbf{b}$ is

$$\left[\begin{array}{ccc|c} 11 & 12 & -7 & 5 \\ 12 & 120 & -84 & 22 \\ -7 & -84 & 59 & -15 \end{array}\right]$$

The reduced row echelon form of this matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{7} & \frac{2}{7} \\ 0 & 1 & -\frac{5}{7} & \frac{13}{84} \\ 0 & 0 & 0 & 0 \end{array}\right]$$

from which it follows that there are infinitely many least squares solutions, and that they are given by the parametric equations

$$\begin{aligned} x_1 &= \tfrac{2}{7} - \tfrac{1}{7}t \\ x_2 &= \tfrac{13}{84} + \tfrac{5}{7}t \\ x_3 &= t \end{aligned}$$

As a check, let us verify that all least squares solutions produce the same least squares error vector and the same least squares error. To see that this is so, we first compute

$$\mathbf{b} - A\mathbf{x} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 & 2 & -1 \\ 1 & -4 & 3 \\ 1 & 10 & -7 \end{bmatrix} \begin{bmatrix} \frac{2}{7} - \frac{1}{7}t \\ \frac{13}{84} + \frac{5}{7}t \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{7}{6} \\ -\frac{1}{3} \\ \frac{11}{6} \end{bmatrix} = \begin{bmatrix} \frac{5}{6} \\ -\frac{5}{3} \\ -\frac{5}{6} \end{bmatrix}$$

Since $\mathbf{b} - A\mathbf{x}$ does not depend on t, all least squares solutions produce the same error vector, namely

$$\|\mathbf{b} - A\mathbf{x}\| = \sqrt{\left(\tfrac{5}{6}\right)^2 + \left(-\tfrac{5}{3}\right)^2 + \left(-\tfrac{5}{6}\right)^2} = \tfrac{5}{6}\sqrt{6} \quad \blacktriangleleft$$

Remark

The normal system is consistent, so there is either exactly one solution or there are infinitely many solutions (depending on whether $A^T A$ is invertible or not). We deduce that the same holds for the least square solutions of $Ax = b$.

A general result

If A is an $m \times n$ matrix, then the following are equivalent:

- The column vectors of A are linearly independent.
- $A^T A$ is invertible.

In particular, if A is a square matrix, then $A^T A$ is invertible if and only if A is invertible.

Proof as exercise.

Unique least square solution

If A is an $m \times n$ matrix with linearly independent column vectors (thus $A^T A$ is invertible), then for every $m \times 1$ matrix b, the linear system $Ax = b$ has a unique least squares solution. This solution is given by

$$x = (A^T A)^{-1} A^T b$$

Consequently $proj_{col(A)} b = Ax = A(A^T A)^{-1} A^T b$

Danger: We cannot simplify the above formula:
$(A^T A)^{-1} = A^{-1}(A^T)^{-1}$ does not make sense if A is not square or not invertible.

For the first example we get

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b} = \begin{bmatrix} 14 & -3 \\ -3 & 21 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$$

$$= \frac{1}{285} \begin{bmatrix} 21 & 3 \\ 3 & 14 \end{bmatrix} \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{17}{95} \\ \frac{143}{285} \end{bmatrix}$$

which agrees with the result obtained in Example 1.

Remark: QR-decomposition and least square solution

If A is an $m \times n$ matrix with linearly independent column vectors, and if $A = QR$ is a QR-decomposition of A, then for each b in $\mathbb{R}^m$ the system $Ax = b$ has a unique least squares solution given by

$$x = R^{-1}Q^T b$$

Proof. Substitute $A = QR$ in

$$x = (A^T A)^{-1} A^T b$$

and make use of $Q^T Q = I$:

$$x = ((QR)^T QR)^{-1}(QR)^T b = (R^T Q^T QR)^{-1} R^T Q^T b$$

$$= (R^T R)^{-1} R^T Q^T b = R^{-1}(R^T)^{-1} R^T Q^T b = R^{-1} Q^T b$$